On Total Isolation in Graphs

Geoffrey Boyer[1], Wayne Goddard[1,2] and Michael A. Henning[2]

[1]School of Mathematical and Statistical Sciences, Clemson University

[2]School of Mathematics and Applied Mathematics, University of Johannesburg

Abstract

An isolating set in a graph is a set S of vertices such that removing S and its neighborhood leaves no edge; it is total isolating if S induces a subgraph with no vertex of degree 0. We show that most graphs have a partition into two disjoint total isolating sets and characterize the exceptions. Using this we show that apart from the 7-cycle, every connected graph of order $n \geq 4$ has a total isolating set of size at most $n/2$, which is best possible.

1 Introduction

We continue the study of versions of isolating set in a graph. Given a family $\mathcal{F}$ of graphs, Caro and Hansberg [6] defined an ***$\mathcal{F}$-isolating set*** of graph G as a set S of vertices such that $G - N[S]$ contains no copy of any graph in $\mathcal{F}$, where $N[S]$ denotes the closed neighborhood of S. The case where $\mathcal{F} = \{K_1\}$ coincides with a dominating set. The case where $\mathcal{F} = \{K_2\}$ is now simply called an ***isolating set***. In [5] it was observed that an isolating set coincides with the concept of vertex-edge dominating set, as introduced by Peters [11] and studied in [4, 10]. Along with the original paper of Caro and Hansberg, several choices for $\mathcal{F}$ have been studied, including where $\mathcal{F}$ is all cycles [1], contains a particular star [15], or contains a particular complete graph [2].

In this paper we instead consider a restriction on the set S itself. Specifically, a ***total isolating set*** is one where S induces a subgraph with no vertex of degree 0. Further, we define the ***total isolation number*** $\iota^t(G)$ of a graph G as the minimum size of such a set. This parameter was originally introduced by Boutrig and Chellali [3] as the ***total vertex-edge domination number***. Amongst other things, they showed that for trees the parameter is at most half

[0]Corresponding author: goddard@clemson.edu

the order, and this is best possible, while in general it is NP-hard to compute. Further bounds for trees are given in [13] and the complexity is further studied in [12].

In this paper we establish a sharp upper bound on the parameter. The sharp upper bound for ordinary isolation was determined by Caro and Hansberg [6] and later Żyliński [14], who showed that apart from the 5-cycle, every graph on $n \geq 3$ vertices has isolation number at most $n/3$. Building on work of Lemańska et al. [9], the extremal graphs were characterized in [5]. We extend here the bound of Boutrig and Chellali [3] and show that, apart from C_7, a graph G of order $n \geq 4$ has $\iota^t(G) \leq n/2$. Our approach is to show that a graph has two almost-disjoint total isolating sets. In this regard it is well-known that a connected nontrivial graph has two disjoint dominating sets, but is not guaranteed to have two disjoint total dominating sets (see [16]). We [5] showed that every connected graph apart from K_2 and C_5 has three disjoint isolating sets.

2 Statements of the Results

For graph G, we define a ***2TI-coloring*** as a coloring (R, B) of the vertices of G with red and blue where each color forms a total isolating set. It is easy to see that if connected G has minimum degree at least 1, then it has a 2TI-coloring if and only it has two disjoint total isolating sets. For a coloring, we say a vertex is ***fully dominated*** if its closed neighborhood contains both colors. So a 2TI-coloring is equivalent to saying that the vertices that are not fully dominated form an independent set.

To characterize graphs with a 2TI-coloring, we need the following definitions. If H is a bipartite graph with bipartition (X, Y), we define a ***half-corona on*** X as the graph obtained from H by adding one or more end-vertices at every vertex of X. The result is a bipartite graph, say with bipartition (X, Y^*). We say that a graph is a half-corona if it arises in this manner. For example, K_2 and P_5 are half-coronas.

We define ***curling*** to mean the following operation: if there is an end-vertex z with neighbor y of degree 2, then add the edge between z and y's other neighbor. For example, this produces K_3 from the half-corona P_3.

We define $\mathcal{H}$ as the set of all graphs that can be obtained as follows:

> *Start with a bipartite graph H that does not contain a cycle of length a multiple of 4, take a half-corona of H, and then do some (possibly empty) set of curlings.*

Figure 1 gives an example of a graph in $\mathcal{H}$, where the hollow vertices form X.

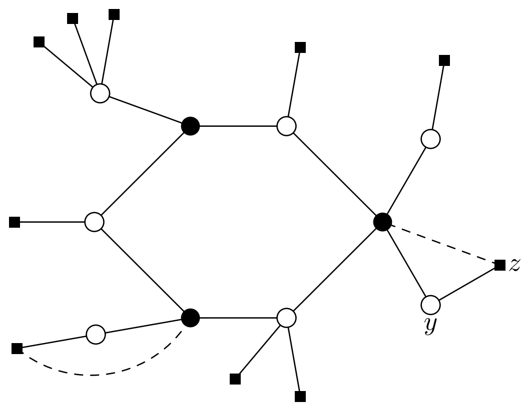

Figure 1: A half-corona with curling

We prove in Section 3:

Theorem 1 *A connected graph on $n \geq 2$ vertices has a 2TI-coloring if and only if it is not the 7-cycle nor in $\mathcal{H}$.*

As a consequence, we obtain a sharp upper bound on the total isolation number of a graph, with proof given in Section 4.

Theorem 2 *If G is a connected graph of order $n \geq 4$, then $\iota^t(G) \leq n/2$ unless G is the 7-cycle.*

As observed above, Boutrig and Chellali [3] showed that the bound $n/2$ is best possible. Specifically, for a graph H they defined the P_3***-corona of*** H by, for every vertex v of H, adding a P_3 and joining one of its ends to v. They showed that for H a tree the total isolation number of the P_3-corona is half its order, but it is immediately true in general. One can further show that curling does not change the total isolation number; and thus some of the P_3's can be replaced by K_3's, and one still has a graph with total isolation number half its order. See Figure 2.

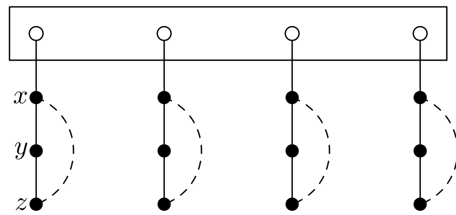

Figure 2: A graph with maximum total isolation number

3 Proof of Theorem 1

3.1 Proof of Necessity

It is easy to show that the 7-cycle has no 2TI-coloring. Recall that a ***support*** vertex is one that has an end-vertex neighbor.

Lemma 3 *If G is a half-corona of bipartite graph H, and G has a $2TI$-coloring, then H has a cycle of length a multiple of* 4.

Proof. Assume G has a 2TI-coloring (R, B) and that G was formed by adding end-vertices to partite set X of H with bipartition (X, Y). Note that each end-vertex of G receives the same color as its support neighbor. Consider any support vertex v_0; say it is red. Note that by construction v_0 is in X.

So that the edge joining v_0 to its end-vertex neighbor is deleted when the blue vertices are removed, the vertex v_0 must have a blue neighbor, say w_0. Further, vertex w_0 must have a blue neighbor, say v_1. This means that $w_0 \in Y$ and $v_1 \in X$. Similarly, v_1 has a red neighbor $w_1 \in Y$ that has a red neighbor $v_2 \in X$.

One can think of this as a walk v_0, w_0, v_1, w_1, v_2 etc. in G. Since the graph is finite, we will eventually reach a repeated vertex, say u. The portion of the walk from the first occurrence of u to its second occurrence is a cycle C. Note that on the cycle, the vertices alternate between X and Y. And if one considers only the vertices of X, they alternate colors. Thus the cycle C has length a multiple of 4. Finally note that all cycles in G are also in H. QED

We need the following lemma about curling.

Lemma 4 *Let G be a graph and let G' be the result of a single curling. Then G' has a 2TI-coloring if and only if G does.*

PROOF. Say the curling adds edge zx where in G vertex z is an end-vertex and y its neighbor. Suppose G has a 2TI-coloring (R, B) with z red. Then y must be red and x must be blue. So this is a 2TI-coloring of G'. Conversely, suppose G' has a 2TI-coloring with z red. Then since both y and z need a neighbor of their color, they must have the same color, while vertex x must be blue. So this is a 2TI-coloring of G. QED

It follows that if G is produced by a set of curlings (possibly none) from a half-corona that has no cycle of length a multiple of 4, then G does not have a 2TI-coloring.

3.2 Proof of Sufficiency

To complete the proof of Theorem 1 we prove the sufficiency. That is, we prove that if connected graph G with order $n \geq 2$ is neither the 7-cycle nor in $\mathcal{H}$, then it has a 2TI-coloring. The proof is by induction on n, and subject to this, by induction on the number of edges of G. Thus by Lemma 4 we may assume that G does not contain a triangle where two of the vertices have degree 2. The base case of the induction is when G is a star, but that is in $\mathcal{H}$.

We will need the following structural idea. Define a ***side-pair*** as two adjacent non-end-vertices s_1 and s_2 such that $G - \{s_1, s_2\}$ is connected. Define a ***side-star*** as a star S such that $G - S$ is connected and all but the center vertex s of S are end-vertices in G.

Lemma 5 *For any connected graph G that is not a star, there exists either a side-pair or a side-star.*

PROOF. Consider a spanning tree T of G and let y be a penultimate vertex on a maximal path of T. Let Z be the set of end-vertex neighbors of y in T. If any two vertices in Z are adjacent in G, then they form a side-pair. So assume Z is an independent set. If Z has more than one vertex, and has a vertex with a neighbor in G other than y, then remove this vertex from Z. Eventually we reach the situation that either only one vertex of Z remains, or every remaining vertex in Z is an end-vertex in G. So y together with the remaining vertices of Z is either a side-pair or a side-star. QED

We resume the proof of sufficiency. We say a connected graph is "good" if it is neither K_1, the 7-cycle nor in $\mathcal{H}$.

Claim 1 *If S is a side-pair or side-star such that $G - S$ is good, then we can induct.*

PROOF. Assume S is a side-pair $\{s_1, s_2\}$. Give $G - S$ a 2TI-coloring by the induction hypothesis. Let f_1 be a neighbor of s_1 and f_2 a neighbor of s_2 in $V(G) - S$, possibly with $f_1 = f_2$. If f_1 and f_2 have the same color, then give both s_1 and s_2 the other color. If f_1 and f_2 have different colors, then give s_1 the color of f_1 and s_2 the color of f_2. In each case, both s_1 and s_2 are fully dominated, and so the result is a 2TI-coloring of G.

Assume S is a side-star with central vertex s. Again give $G - S$ a 2TI-coloring by the induction hypothesis. Then give all of S the opposite color to the neighbor of s in $V(G) - S$. Vertex s is fully dominated and the end-vertices of S have the same color as their neighbor. So the result is a 2TI-coloring of G. QED

Claim 2 *If G contains a cycle C of length a multiple of 4, then we can induct.*

PROOF. Assume $G - C$ contains an edge. Then in the proof of Lemma 5, one can choose the spanning tree T of G with a maximal path where the penultimate vertex y and Z are disjoint from C (think of shrinking C to a single vertex). So the resultant side-pair/side-star S is disjoint from C. Thus $G - S$ contains a cycle of length a multiple of 4 and hence is good, and we can induct by Claim 1.

Assume $G - C$ contains no edge. Then color the cycle C in pairs, two red, two blue etc. All of C is fully dominated and so this coloring can be extended to a 2TI-coloring of G. QED

So assume G does not contain a cycle of length a multiple of 4.

Claim 3 *If G contains a triangle, then we can induct.*

Proof. Consider a triangle C. By the lack of curlings, we may assume at least two vertices of C have neighbors outside C. If $G - C$ has an edge neither end of which is adjacent to C, then (as in the previous claim) we can choose the side-star/side-pair S so that it does not contain a neighbor of C. So C cannot be a curling in $G - S$ and hence $G - S$ is good, and we proceed as before.

So assume every edge of G has at least one end adjacent to C. In particular, the vertices not dominated by C form an independent set and each has a neighbor adjacent to C. By the lack of 4-cycles, no vertex outside C has two neighbors in C. Say C has vertices v_1, v_2, v_3. There are two cases.

(i) Assume there is some vertex on C, say v_1, that has a neighbor outside C but all such neighbors are end-vertices. Then color all vertices blue except for v_2 and v_3 and any end-vertex neighbors they have. One can readily check that every vertex has a neighbor of its own color. Furthermore, the subgraph $G - N[B]$ contains the end-vertex neighbors of v_2 and v_3; and the subgraph $G - N[R]$ contains the end-vertex neighbors of v_1 as well as vertices not dominated by C. So this is a 2TI-coloring.

(ii) Assume every vertex on C that has a neighbor outside C has one that is not an end-vertex. By the lack of curlings, at least two vertices of C have neighbors outside C, say v_2 and v_3. Then color red all vertices on C and any end-vertex neighbors they have, and color blue all other vertices. One can readily check that every vertex has a neighbor of its own color. Furthermore, the subgraph $G - N[B]$ contains the end-vertex neighbors of v_2 and v_3 as well as either v_1 or end-vertex neighbors of v_1 but not both, since if v_1 has a neighbor outside C at least one of them is blue. And the subgraph $G - N[R]$ contains the vertices not dominated by C. So this is a 2TI-coloring. QED

So assume G contains no triangle.

Claim 4 *If G contains an odd cycle, then we can induct.*

Proof. Let C be a shortest odd cycle, necessarily induced. If $G-C$ has an edge, then as before we can choose the side-star/side-pair S to be disjoint from C. Thus $G-S$ contains an odd cycle, and is therefore good except when it is exactly C_7. It is easy, if a bit tedious, to show that if $G-S$ is a C_7 then G has a 2TI-coloring; we omit the details.

So assume $G-C$ has no edge. By the minimality of C and the lack of 3- and 4-cycles, it follows that no vertex outside C has two neighbors in C. (Suppose vertices v and v' on C have common neighbor w: if v and v' are consecutive, this creates a triangle; if they are at distance 2, this creates a 4-cycle; and if they are at distance 3 or more, one can replace one of the segments of C with vwv' to create a shorter odd cycle.) That is, G is obtained from the cycle C by adding some end-vertices with neighbors on the cycle.

If G is not a cycle, then we can choose a side-star S_v consisting of vertex v on C with its neighbors outside C. If $G-S_v$ is in $\mathcal{H}$, then since v's neighbors on C are in different partite sets of bipartite $G-S$, exactly one of them must have a neighbor outside C and every alternate vertex on C must be a support vertex. But then consider the support vertex w at distance two from v, and the side-star S_w consisting of w and its neighbors outside C. Vertex w does not have a neighbor that is a support vertex, and so $G-S_w$ is not in $\mathcal{H}$; that is, it is good.

If G is a cycle, then a pair of consecutive vertices form a side-pair S. If $G-S$ is not good, then it must be a path on three or five vertices. This means G is the 5- or 7-cycle. But it is easily seen that the 5-cycle has a 2TI-coloring. QED

So we may assume G is bipartite.

We will need the following "almost" TI-coloring. For vertex f, define an ***f-TAI-coloring*** (total almost-isolating) as one where every vertex has a neighbor of its color, vertex f is red, and the set of vertices that are not fully dominated induces a graph whose only possible nontrivial component is a red star centered at f.

Lemma 6 *If graph F is a half-corona of order at least 3 with bipartition (X, Y^*), then for all $f \in X$ there is an f-TAI coloring.*

Proof. Color each vertex based on its distance from f: vertices at distance $0, 1, 4, 5, 8, 9$ etc get red, and the remainder get blue. Note that every vertex of X is a support vertex and hence has an end-vertex neighbor that is farther from f. It follows that every vertex in X, except f, has neighbors of both colors. Further, every vertex of Y^* is at odd distance from f and so has the same color as its neighbor(s) nearer to f. It follows that the vertices that are not fully dominated are a subset of $\{f\} \cup Y^*$. That is, this is a f-TAI-coloring. QED

Claim 5 *If G is bipartite, then we can induct.*

Proof. Consider the side-star/side-pair S guaranteed by Lemma 5. If $G - S$ is good, then we proceed as before. So assume $G - S$ is not good. If $G - S$ is K_1, then G is a star, a contradiction. If $G - S$ is K_2, then color $G - S$ red and S blue. So assume $G - S$ is a half-corona with all vertices in X support vertices.

If S is a side-star, then s must have neighbor f in X, since otherwise G is in $\mathcal{H}$. So give $G - S$ the f-TAI coloring, with f red, and color all of S blue. This is a 2TI-coloring of G.

Finally suppose S is a side-pair. Let f_1 be a neighbor of s_1 and f_2 a neighbor of s_2 in $V(G) - S$, possibly with $f_1 = f_2$. By the lack of odd cycles and cycles of length a multiple of 4 in G, the distance between f_1 and f_2 in $G - S$ is congruent to 1 mod 4. Say f_1 is in X. Then do the f_1-TAI coloring of $G - S$ as in the above lemma, with f_1 red, and note that by construction, vertex f_2 is also red. So color S blue and we have a 2TI-coloring of G. QED

This completes the proof of Theorem 1.

4 Proof of Theorem 2

It is easy to see that $\iota^t(C_7) = 4$. So assume G is not C_7. If G has a 2TI-coloring then the bound $\iota^t(G) \leq n/2$ is immediate. If G is obtained from a star by a (possibly empty) set of curlings, then its total isolation number is 2. So assume G is obtained by a (possibly empty) set of curlings of a half-corona G' that is not a star.

Let x be a support vertex of G' and let J be the graph formed from G' by removing all end-vertex neighbors of x. Since J is not a half-corona and not C_7, it has a 2TI-coloring; say (R, B) where $x \in R$. Then R is a total isolating set of G' and hence G. If B is not a total isolating set of G', then all neighbors of x in J are in R. But if y is such a neighbor, it must have a neighbor in B, and so $B \cup \{y\}$ is a total isolating set of G' and hence G. That is, we have two total isolating sets of G that overlap in at most one vertex and neither contains the end-vertex neighbor(s) of x. The bound follows by averaging.

5 Further Partitioning Problems

In this section, we consider two related partitioning problems.

In contrast to Theorem 1 (and the case for ordinary isolating sets [5]), we show that the question of whether a graph has three disjoint total isolating sets is NP-complete. We reduce from partitioning into independent dominating sets, which is known to be NP-complete [7, 8]. Let $S(G)$ denote the subdivision graph of G, and let $C(H)$ denote the corona of graph H formed by adding one end-vertex adjacent to every vertex of H. Then:

Lemma 7 *For graph G, the graph $C(S(G))$ has three disjoint total isolating sets if and only if G has a proper 3-coloring where every vertex has neighbors of both other colors.*

Proof. If G has such 3-coloring, then color $C(S(G))$ as follows: give every vertex of $V(G)$ its color in G; for each subdivision vertex s_{uv} give it the color missing from its two neighbors u and v; and for each end-vertex give it the color of its neighbor. Every vertex of $S(G)$ has a neighbor of its color. The coloring of a subdivision vertex is chosen so that it is fully dominated. Consider a vertex $u \in V(G)$ in $C(S(G))$. For each color different from it, say blue, there is a vertex $v \in V(G)$ that is a neighbor in G and is also not blue; so s_{uv} has color blue. It follows that all of $V(S(G))$ is fully dominated. That is, the vertices of $C(S(G))$ that are not fully dominated are the end-vertices, and they form an independent set.

Conversely, consider a 3-coloring of $C(S(G))$ into total isolating sets. In order for the leaf-edge incident to a subdivision vertex s_{uv} to be ve-dominated by all three colors, the three vertices s_{uv}, u, and v must all have different colors. So the coloring restricted to G is a proper coloring. In order for the leaf-edge incident to a vertex $u \in V(G)$ to be ve-dominated by all three colors in $C(S(G))$, vertex u must have a subdivision neighbor of both other colors, which means that for both other colors there must be a vertex $v \in V(G)$ that is a neighbor in G and has that color. That is, the coloring restricted to G has the desired property. QED

One can also think of graphs with two disjoint total dominating sets. It is trivial that a tree does not have two disjoint total dominating sets, since each total dominating set must contain all support vertices. But one can ask for something weaker. We define a ***2TAD-coloring*** (total almost dominating) of a tree as a 2-coloring where every non-end-vertex has neighbors of both colors and every end-vertex has a neighbor of their color. Note that this is a stronger condition than 2TI-coloring.

Theorem 8 *A tree on at least* 3 *vertices has a* 2*TAD-coloring if and only if there exist end-vertices in both partite sets.*

PROOF. (1) Suppose there is a 2TAD-coloring. Let u_0 be an end-vertex. Build a maximal path according to the following rule: for odd i let u_i be a neighbor of u_{i-1} of the same color, and for even i let u_i be a neighbor of u_{i-1} of the other color. Since every vertex has a neighbor of its own color, the process will terminate with i even and some u_{i-1} does not have a neighbor of the other color. It follows that u_{i-1} is an end-vertex, and at odd distance from u_0.

(2) Suppose there exist end-vertices in both partite sets. Then the path P joining them contains an even number of vertices. Color this path two red, two blue, two red, etc. Every vertex on P except the ends is fully dominated.

Assume there remains an uncolored support vertex. Since the colored vertices form a connected subgraph, there must be a nontrivial path Q of uncolored vertices starting from a vertex that has a colored neighbor and ending at an end-vertex. Say Q starts with vertex x that has red neighbor y. Then if Q has an even number of vertices, color x and the second vertex of Q blue, the next two red, and so on. If Q has an odd number of vertices, color x red, the second and third vertex of Q blue, and so on. After this additional coloring it still holds that the colored vertices form a connected subgraph, all colored non-end-vertices have neighbors of both colors, and all colored end-vertices have the same color as their neighbor. Continue until all support vertices are colored.

Now, consider any uncolored vertex u. Since the colored vertices form a connected subgraph, there must be an uncolored end-vertex reachable from u. But since all support vertices are colored, this means u is an end-vertex. So give each uncolored end-vertex the color of its neighbor, and the result is a 2TAD-coloring. QED

6 Conclusion

The most obvious open problem is to determine the graphs where the total isolation number is half the order. We noted that the P_3-coronas and curling thereof (all of which have order a multiple of 4) are examples. Maybe apart from this general family there are only finitely many extremal graphs. For $n = 6$ the only example is C_6. For $n = 8$, apart from the general family, there is C_8 and four graphs obtained from C_7 by adding one vertex. Computer search shows there is no example of order 10 at all. For $n = 12$, the cycle C_{12} is an example not in the general family, but we do not know the full list.

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